A perfect even tuple dominating the halved coordinates of a bounded even tuple

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Editorial note. The counted unit is the article's own Lemma l:first at source lines 154-156 together with the proof environment that begins on the next line (157-170); the pairing is the ordinary adjacent one and the proof ends before the author's proof of Theorem t:perfect opens at source line 172.

Editorial note. The retained context at source lines 30-36 and 51 is the article's own statement of the Balister-Gyori-Schelp necessary condition and conjecture and the article's own definitions of profile, admissible, even and perfect tuples.

Editorial note. The retained context at source line 64 is the article's own definition of the precedence order and of the concatenation of tuples; it is retained because the counted lemma concludes y <= y' and because the down-closedness of admissible tuples stated there is what makes the dominating tuple useful.

Editorial note. The retained context at source line 152 is the article's own definition of the quantity o(x); it is retained because the hypothesis x_{n+1} >= 2 o(x) of the counted lemma is stated in terms of it.

Editorial note. The native preamble is retained from source lines 2-18, that is the article class, geometry, AMS, bm, mathtools, comment, enumerate and tikz packages and the author's theorem declarations; the title and author lines at source lines 19-20 are not part of the preamble fragment and the document body is not retained.

Editorial note. No retained line contains a citation, so the material carries no bibliography; the retained lines refer only to the label c:BGS, which the retained conjecture at source lines 34-36 defines, and to the label t:perfect, which is left undefined by the retained fragments and is therefore declared as a non-load-bearing navigation pointer rather than as a premise.

Editorial note. The article's Theorem t:perfect, Lemma l:admissible and discussion theorem c:admissible are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Profiles of matchings in the hypercube Q^n. For a matching M in the cube Q^n the profile is the vector x in N^n whose i-th coordinate counts the edges of M whose endpoints differ in the i-th coordinate; a tuple is admissible when it is the profile of some matching, even when every coordinate is even, and perfect when the sum of the coordinates equals 2^{n-1}; the article's own definitions are retained at source line 51.. Fixes the meaning of admissible, even and perfect for the counted lemma, whose conclusion asserts the existence of a perfect even tuple in N^n.

- Precedence order on tuples and down-closedness of admissible tuples. For x and z in N^n the article writes z precedes x, in symbols z <= x, when z_i <= x_i for every i; the set of admissible tuples is down-closed in this partial order, and (x,z) denotes the concatenation of x in N^n and z in N^m as a vector in N^{n+m}; retained at source line 64.. The counted lemma concludes y <= y' for a suitable perfect even y' in N^n, which is exactly this order; the down-closedness is why a perfect even tuple dominating y transfers admissibility to y.
- The parity obstruction of Balister, Gyori and Schelp. If the cube F_2^n is partitioned into pairs {a_i,b_i} with differences d_i, then the sum of all d_i vanishes, because each vertex occurs exactly once on each side of the sum; the article records this identity at source lines 30-33 and the conjecture that it is also sufficient at source lines 34-36.. Background for the evenness condition that the counted lemma assumes on its input tuple x in N^{n+1}.

Source reference. Original theorem 1.2: sources/combinatorics-fl03-matchings-in-the-hypercube-with-sp/pinned-original-source.tex, source lines 53–55. The retained context sentence at source line 64 names Theorem t:perfect when it records that the equivalence of admissibility and evenness for perfect tuples is the main theorem. The counted lemma l:first and its proof never use that theorem; the name is a navigation pointer only.

$$\sum_{i=1}^{2^{n-1}} \boldsymbol{d_i} = \sum_{i=1}^{2^{n-1}} \boldsymbol{a_i} - \boldsymbol{b_i} = \sum_{i=1}^{2^{n-1}} \boldsymbol{a_i} + \boldsymbol{b_i} = \sum_{\boldsymbol{v}\in\mathbb{F}_2^n} \boldsymbol{v} = \boldsymbol{0}. \tag{1}$$

They conjectured that this necessary condition was also sufficient.

Conjecture 0.1 (Balister, Győri, and Schelp)**.** *Let $n \geq 2$ and let $\boldsymbol{d_1}, \ldots, \boldsymbol{d_{2^{n-1}}}$ be elements of $\mathbb{F}_2^n$ such that $\sum_{i=1}^{2^{n-1}} \boldsymbol{d_i} = \boldsymbol{0}$. Then there exists a partition of $\mathbb{F}_2^n$ into pairs $\{\boldsymbol{a_i}, \boldsymbol{b_i}\}$ such that $\boldsymbol{a_i} - \boldsymbol{b_i} = \boldsymbol{d_i}$ for each $i \in \left[2^{n-1}\right]$.*

Given a matching $M \subseteq Q^n$ we say the *profile* of M is the vector $\boldsymbol{x} = (x_1, \ldots, x_n) \in \mathbb{N}^n$ such that M contains x_i edges whose endpoints differ in the ith coordinate, and we are interested in which tuples $\boldsymbol{x} \in \mathbb{N}^n$ are achievable as the profile of some matching. We call such tuples *admissible*. In this case, the obvious necessary condition for a tuple to be the profile of a perfect matching is that each unit vector appears an even number of times as a matching edge, i.e., each x_i is even, in which case we say that $\boldsymbol{x}$ is *even*, and Conjecture 0.1 would imply that this condition is also sufficient. The aim of this short note is to verify the conjecture in this case. Let us call a tuple $\boldsymbol{x} \in \mathbb{N}^n$ *perfect* if $||\boldsymbol{x}||_1 := \sum_i x_i = 2^{n-1}$.

We first note that by symmetry the ordering of the coordinates of $\boldsymbol{x}$ is irrelevant, and so we may assume from this point forwards that the coordinates appear in non-decreasing order, i.e., $x_1 \leq x_2 \leq \ldots \leq x_n$. Given $\boldsymbol{x}, \boldsymbol{z} \in \mathbb{N}^n$ we say that $\boldsymbol{z}$ *precedes* $\boldsymbol{x}$, which we write as $\boldsymbol{z} \preceq \boldsymbol{x}$, if $z_i \leq x_i$ for all $i \in [n]$. Clearly the set of admissible tuples is down-closed in the partial order $\preceq$. In particular, Theorem 1.2 is equivalent to the statement that every even tuple is admissible. Finally, given $\boldsymbol{x} \in \mathbb{N}^n$ and $\boldsymbol{z} \in \mathbb{N}^m$ we will write $(\boldsymbol{x}, \boldsymbol{z})$ for the vector in $\mathbb{N}^{n+m}$ obtained by concatenation $\boldsymbol{x}$ and $\boldsymbol{z}$.

Let us make the preceding idea formal with the following lemma. Given $\boldsymbol{x} \in \mathbb{N}^n$, let us write $o(\boldsymbol{x})$ for the number of coordinates of $\boldsymbol{x}$ except the last which are congruent to 2 mod 4.

Lemma 0.2. *Let $n \geq 2$, and let $\boldsymbol{x} \in \mathbb{N}^{n+1}$ be even with $||\boldsymbol{x}||_1 \leq 2^n$ and $x_{n+1} \geq 2o(\boldsymbol{x})$. If $\boldsymbol{y} := \left(\frac{x_1}{2}, \frac{x_2}{2}, \ldots, \frac{x_n}{2}\right)$, then there exists a perfect even $\boldsymbol{y'} \in \mathbb{N}^n$ such that $\boldsymbol{y} \preceq \boldsymbol{y'}$*

Proof. For each $i \in [n]$ let

$$o_i = \begin{cases} 1 & \text{if } x_i \equiv 2 \mod 4, \\ 0 & \text{otherwise.} \end{cases}$$

Let $\boldsymbol{z} = \boldsymbol{y} + \boldsymbol{o}$. Note, in particular, that $\boldsymbol{y} \preceq \boldsymbol{z}$ and that by construction $\boldsymbol{z}$ is even.

Since $||\boldsymbol{o}||_1 = o(\boldsymbol{x}) \leq \frac{x_{n+1}}{2}$, it follows that

$$||\boldsymbol{z}|| \leq \sum_{i=1}^{n} \frac{x_i}{2} + o_i \leq \frac{||\boldsymbol{x}||_1}{2} \leq 2^{n-1}.$$

Hence, since 2^{n-1} is even, there is some perfect even $\boldsymbol{y'} \succeq \boldsymbol{z} \succeq \boldsymbol{y}$ as claimed. □